



Factors associated with high healthcare resource utilisation among COPD patients

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Received 4 June 2012; accepted 12 September 2012

Available online 9 October 2012

KEYWORDS

COPD;
Hospitalisation;

Summary

Background: Chronic obstructive pulmonary disease (COPD) places a huge economic burden on healthcare systems, especially patients with frequent exacerbations and co-morbidities.

Abbreviations: BMI, body mass index; COPD, Chronic Obstructive Pulmonary Disease; CRP, C-reactive protein; FEV₁, forced expiratory volume in one second; FVC, forced vital capacity; EQ-5D, EuroQol five-dimension questionnaire; GOLD, Global Initiative for Chronic Obstructive Lung Disease; GP, General practitioner; HADS, Hospital Anxiety and Depression Scale; ICU, Intensive care unit; LCDAL, London Chest Activity of Daily Living Scale; LTOT, long-term oxygen therapy; MRC, Medical Research Council; SABAs, short-acting beta-2 agonists; 6MWT, six-minute walk test.

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Physical activity;
Healthcare resource
utilisation;
Systemic
inflammation

Objectives: To identify factors associated with high utilisation of healthcare resources in a population of patients with COPD.

Method: We conducted an observational, cross-sectional, multicentre study with the aim of identifying the factors associated with high resource utilisation among patients with COPD. Sociodemographic and anthropometric characteristics of the study population, as well as data on health-related quality of life, respiratory symptoms, presence of anxiety and depression, physical activity and lung function were collected. We examined the relationship between these variables and high utilisation of healthcare resources, by performing a multivariate analysis based on a logistic regression model.

Results: 115 patients (64 were high users of healthcare resources, and 51 control patients) from 13 hospitals were selected. Patients presenting high resource utilisation had worse FEV₁, worse basal SpO₂, less distance walked in the 6-minute walk test, and increased dyspnoea. They also had a worse BODE index, worse scores in all dimensions of the EURO-QOL 5D and the LCADL scale, and displayed a higher prevalence of depression. Multivariate analysis yielded a statistically significant association between SpO₂, LCADL scores, serum fibrinogen values and total leukocyte count, and high healthcare resource utilisation.

Conclusions: COPD patients who incur higher healthcare resource utilisation show reduced physical activity, increased respiratory failure and increased systemic inflammation.

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Introduction

COPD is currently the fourth leading cause of mortality throughout the world and it is estimated to be the third cause of death by the year 2020. Given its high prevalence and chronicity, COPD is a social burden and places substantial healthcare costs on public health systems.^{1–3} This economic burden increases with disease progression, age and frequent exacerbations, which might represent up to 70% of all health expenditure and have an increasingly demonstrated relationship with reduced survival.³

When it comes to assessing COPD severity, besides the predictive ability of FEV₁, there are various approaches that are used to improve prediction of mortality in COPD patients. In fact, the multidimensional BODE index (body mass index, obstruction, dyspnoea and exercise capacity) appears to be a better predictor of mortality due to respiratory causes⁴ than FEV₁. Assessment and scoring of dyspnoea and health status by means of specific questionnaires also provide useful information on the impact of exacerbations and the efficacy of pharmacological and non-pharmacological therapy such as pulmonary rehabilitation.⁵ There is growing evidence of the systemic effects of COPD, namely systemic inflammation, skeletal muscle dysfunction, hormone alterations, cardiovascular effects and osteoskeletal effects.^{6,7} Malnourishment, sociocultural and economic background, and psychological disorders, such as anxiety and depression, are other factors that also influence the health status of COPD patients and may increase the risk for hospitalisation due to exacerbations.^{8,9}

In daily clinical practice, it is not uncommon to find COPD patients who, regardless of COPD severity, repeatedly require healthcare assistance; not only in terms of hospitalisations but also scheduled and non-scheduled office visits and emergency visits. There are a number of previous studies that have tried to characterise the aforementioned group of patients with high healthcare resource utilisation in order to identify possible ways of action to

manage such individuals.^{10–12} Notwithstanding, the outcomes yielded differ across the studies. In this study, we collected data from a series of tests performed in a group of mild to severe COPD patients monitored by the respiratory services of Spanish public hospitals, in order to identify the clinical and functional characteristics associated with the high utilisation of healthcare services.

Method

The present study was a multicentre, observational study aimed at identifying the characteristics of COPD patients associated with high utilisation of healthcare resources. The characteristics of this study have been described in a previous investigation examining hospital admissions for exacerbations.¹³ In the current study we defined the concept of high healthcare resource utilisation and analysed the determinants associated with this type of patients.

The inclusion criteria were patients aged >40 yrs. with physician-diagnosed COPD according to the GOLD criteria (post-bronchodilator FEV₁/FVC < 0.7), were clinically stable (no exacerbations in the last four weeks), and were current or former smokers with a smoking history of ≥10 pack-yrs. High resource utilisation was defined as those patients who, in the year prior to their participation in the study, had at least: a) 1 hospital admission for COPD exacerbation, b) 2 or more visits to emergency units due to an exacerbation, or c) 2 or more non-scheduled COPD-related visits to outpatient clinics. The remaining patients were defined as patients with low resource utilisation or control group.

Data were obtained by means of a clinical interview including demographic characteristics and socioeconomic and education background. The number of COPD-related visits for medical assistance in the year prior to the study was also collected. Moreover, data obtained from routine

tests were examined, namely haemogram, biochemical tests for ferritin levels, fibrinogen, total protein, α -1 antitrypsin plasma levels and arterial blood gas measurement taken of room air. Pulmonary function testing was also performed in clinically stable patients, by means of spirometry and a post-bronchodilator test. A 6-min walking test was carried out, following the SEPAR (Spanish Society of Pneumology and Thoracic Surgery) guidelines.¹⁴ The BODE index was estimated as described elsewhere.⁴ The total BODE score may range from 0 to 10 points; the higher the BODE score, the higher the risk of death.

Comorbidity was assessed with the Charlson Comorbidity Index.¹⁵ Whenever possible, an electrocardiogram (ECG) and transthoracic echocardiography were performed to test heart function.

Health-related quality of life (HRQoL) was measured with the European quality of life (EuroQoL) five-dimension questionnaire (EQ-5D).¹⁶ The EQ-5D is a standardised generic instrument which is applied in a variety of diseases and provides useful information through a self-classifier/descriptive system of the respondent's own health and a single summary index value representing his/her overall health state. The questionnaire consists of five questions that are converted into a single summary index (the VAS scores range from 0 to 100, and scores close to 100 reflect better health state), and a visual analogue scale (VAS). Dyspnoea was measured by the Medical Research Council (MRC) scale,¹⁷ whereas physical activity was assessed using the validated Spanish version¹⁹ of the London Chest Activity of Daily Living (LCADL) Scale.¹⁸ The LCADL scale is a self-administered 15-item questionnaire consisting of 4 domains or subscales. For each activity the patient reports a score on a rating scale ranging from 0 to 5 (0 = 'I wouldn't do it anyway', and 5 = 'someone else does this for me (or helps)'). The score of each of the 4 domains and the total LCADL score, the latter ranging from 0 (reflecting low disability) to 75 (maximal disability), are then calculated. Patients were also asked to complete the Hospital Anxiety and Depression Scale (HADS),²⁰ a validated tool used to assess symptoms and identify cases of anxiety disorders and depression in patients with chronic disease. This measure comprises 7 depression items (Depression sub-scale or HAD-D) and 7 anxiety items (Anxiety sub-scale or HAD-A). Scores on each sub-scale can be interpreted in ranges from 0 to 21. A score from 0 to 7 indicates no depression or anxiety, a score from 8 to 10 is suggestive of the presence of the respective mood disorder, and a score of 11 or higher probable presence of the mood disorder.

The study protocol was approved by the Ethics Committee of the University Hospital Puerta del Mar (Cádiz, Spain), and all study participants provided written informed consent.

Statistical analysis

The Student's *t* test was used for comparison of normally distributed continuous variables whereas the Mann-Whitney *U* test was used for not normally distributed continuous variables. Chi-square test was applied for comparison of categorical variables and the corresponding *p*-value obtained is presented in the study.

With regard to the assessment of physical activity, a linear regression model, which included the LCADL score as a dependent variable and the distance walked in the 6MWT as the independent variable, together with data adjustment and estimation of the goodness of fit of the model were carried out.

A multivariate analysis based on a logistic regression model was carried out by adjusting the variable under study, namely high resource utilisation, for all the variables showing significant differences across the two study groups in the bivariate analysis. A predictive model for high utilisation of healthcare resources was constructed with the regression analysis variables associated with elevated utilisation of healthcare resources. Therefore, the area under the curve (AUC) and its confidence interval value (95%) were calculated. A *p*-value less than 0.05 was considered to be statistically significant. The data analysis was performed using the SPSS v18 statistical package.

Results

A total of 137 patients were recruited and 115 completed the necessary data to be included in the study; of these 64 were high users of healthcare resources and 51 control participants. The mean age of the study population ($\pm$ DE) was 66.9 ± 8.7 . 93% of the study population were male, presented a smoking history of 51.5 ± 27 pack-years, showed a mean post-bronchodilator FEV₁ (of the predicted value, %p) of $43.7 \pm 15\%$, and had a mean body mass index (BMI) of 26.8 ± 5.6 . The Charlson Comorbidity Index score was 4.4 and the mean BODE index was 3.7. Anthropometric and sociodemographic characteristics, pulmonary and heart function tests, treatment received and biochemical parameters of the study population are displayed in Table 1.

Patients with high utilisation of healthcare resources showed a higher MRC dyspnoea degree (2.03 ± 0.9 versus 1.4 ± 1 ; *p* = 0.001), lower SpO₂ ($89.7\% \pm 12.2$ versus $94.1\% \pm 3.4$; *p* = 0.007) and greater airflow obstruction (FEV₁ $39.6\% \pm 17.6$ versus $46.9\% \pm 16.5$; *p* = 0.024). Moreover, in the case of patients presenting high utilisation of healthcare resources, the distance walked in the 6MWT was shorter ($357.5 \text{ m} \pm 128.4$ versus $425.6 \text{ m} \pm 125.8$; *p* = 0.006), and the total BODE index score was higher (4.3 ± 2.3 versus 3.1 ± 2.1 ; *p* = 0.024).

With respect to the Charlson Comorbidity Index, there were no differences across the study groups, whereas the echocardiographic study performed in a subset of 73 patients (42 of these patients were high users of resources) (Table 2) yielded differences across different parameters. The differences, however, were only significant for the left ventricular ejection fraction, which was lower in the group of high resource utilisation ($59.5\% \pm 9.8$ versus $64.4\% \pm 9.2$; *p* = 0.033). Blood inflammatory parameters were significantly higher in patients with high utilisation of resources: with fibrinogen values of $408.3 \text{ mg/dL} \pm 207$ versus 306.4 ± 192 ; *p* = 0.007 and leukocyte counts of $8.870 \text{ cells/mm}^3 \pm 2.930$ versus $7.320 \text{ cells/mm}^3 \pm 2.036$; *p* = 0.001 (Table 2).

With regard to the health-related quality of life, physical activity, and anxiety/depression questionnaires, significant differences were observed across both study groups

Table 1 Characteristics of the study population.

	Global (n = 115)	High resource utilisation (n = 64, 55.6%)	Control group (n = 51, 44.4%)	p-Value
Age (years)	66.9 ± 8.7	66.6 ± 9.4	67.2 ± 8.0	0.74
Sex (% male)	107 (93.0%)	60 (93.8%)	47 (92.2%)	0.51
Marital status				0.86
Single	11 (9.6%)	7 (10.9%)	4 (7.8%)	
Couples	77 (67.0%)	42 (65.6%)	35 (68.6%)	
With family	27 (23.5%)	15 (23.4%)	12 (23.5%)	
Education background				0.17
W/o education	31 (27.0%)	19 (29.7%)	12 (23.5%)	
Primary	49 (42.6%)	30 (46.9%)	19 (36.3%)	
Secondary	21 (18.3%)	10 (15.6%)	11 (21.6%)	
University	13 (11.5%)	4 (6.3%)	9 (17.6%)	
Missing	1 (0.6%)	1 (1.5%)	0	
Current smokers (%)	28 (24.3%)	15 (23.4%)	13 (25.5%)	0.95
Former smokers (%)	87 (75.7%)	49 (76.6%)	38 (74.5%)	
Pack-years	51.5 ± 27.6	51.0 ± 31.1	52.0 ± 22.3	0.85
BMI (kg/m ²)	26.8 ± 5.8	26.3 ± 6.0	27.4 ± 5.5	0.34
Charlson Index	4.39 ± 2.1	4.19 ± 1.9	4.65 ± 2.4	0.15
Dyspnoea (MRC)	1.75 ± 1.04	2.03 ± 0.9	1.39 ± 1.0	0.001
Spirometry				
FEV ₁ (%)	43.7 ± 15.1	39.6 ± 17.6	46.9 ± 16.5	0.024
FEV ₁ mL	1.205 ± 511	1.058 ± 509	1.310 ± 520	0.010
FVC (%)	69.5 ± 21.5	64.5 ± 26.5	74.0 ± 18.81	0.026
FVC mL	2.560 ± 932	2.344 ± 1144	2.771 ± 887	0.032
FEV ₁ /FVC	46.4 ± 12.8	45.5 ± 14.1	47.1 ± 11.4	0.51
Six-minute walk test, m	386 ± 131	357 ± 128	425 ± 125	0.006
BODE index	3.87 ± 2.30	4.31 ± 2.34	3.13 ± 2.14	0.024
ABG, room air				
PaO ₂ , mm Hg	67.4 ± 12.1	64.4 ± 11.5	71.8 ± 11.8	0.003
PaCO ₂ , mm Hg	44.3 ± 8.7	45.2 ± 8.19	43.0 ± 9.5	0.24
pH	7.42 ± 6.5	7.42 ± 5.12	7.41 ± 8.34	0.50
SpO ₂ %	92.6 ± 4.1	89.7 ± 2.23	94.1 ± 3.45	0.007
Treatment for COPD				
SABA %	93 (80.9%)	57 (89.1%)	36 (70.6%)	0.029
LABA %	103 (89.6%)	58 (90.6%)	45 (88.2%)	0.53
Ipratropium %	9 (7.8%)	5 (7.8%)	4 (7.8%)	0.20
Tiotropium %	99 (86.1%)	57 (89.1%)	42 (82.4%)	0.39
Theophyllines %	21 (18.3%)	12 (18.8%)	9 (17.6%)	0.16
Inhaled corticosteroids %	104 (90.4%)	59 (92.2%)	45 (88.2%)	0.41
Mucolytics %	45 (39.1%)	29 (45.3%)	16 (31.4%)	0.14
LTOT %	25 (21.7%)	21 (32.8%)	4 (7.8%)	0.003
Rehabilitation %	14 (12.2%)	8 (12.5%)	6 (11.8%)	0.27
Antinfluenza vaccination %	83 (72.2%)	48(75%)	35 (68.6%)	0.43
Antipneumococcal vaccination %	40 (34.8%)	18 (28.1%)	22 (43.1%)	0.05

Bold values signify the comparisons with statistically significant p values.

BMI: body mass index. PaO₂: oxygen partial pressure. PaCO₂: carbon dioxide, partial pressure. SpO₂: haemoglobin, oxygen saturation. ABG: arterial blood gases. SABA: short acting beta agonists. LABA: long acting beta agonists. LTOT: long term oxygen therapy.

(Table 3). Patients with high resource utilisation showed significantly worse EQ-5D scores, in terms of total EQ-5D score (0.64 ± 0.35 versus 0.82 ± 0.22 ; $p = 0.002$), VAS and each of its individual domains. Regarding the LCDAL we

found significant differences in the total LCDAL score (47.1 ± 12.4 versus 39 ± 14.1 ; $p = 0.002$) as well as in the self-care and physical domains. The total LCADL score displayed a moderate and statistically significant correlation

Table 2 Biochemical and echocardiographic parameters.

Biochemical	Global (n = 115)	High resource utilisation (n = 64)	Control group (n = 51)	p-Value
Haemoglobin (gr/dL)	14.2 ± 2.7	13.9 ± 2.9	14.5 ± 2.5	0.25
Leukocyte count (10 ³ /μL)	8.190 ± 2.68	8.870 ± 2.93	7.320 ± 2.03	0.001
Fibrinogen (mg/dL)	366.8 ± 204	408.2 ± 207.16	306.1 ± 192.4	0.007
Proteins (mg/dL)	7.18 ± 0.86	7.09 ± 0.92	7.29 ± 0.78	0.053
Alfa1- antitrypsin (mg/dL)	125.3 ± 60.5	118.9 ± 62.65	133.3 ± 57.36	0.20
Ferritin (mg/dL)	152 ± 110	163 ± 129	143 ± 92	0.62
C-reactive protein (mg/dl)	3.16 ± 5.01	3.27 ± 5.33	3.04 ± 6.01	0.86
Echocardiography	Global (n = 68)	High resource utilisation (n = 42)	Control group (n = 26)	p-Value
Right chamber enlargement %	11 (16.1%)	8 (19%)	3 (9.1%)	0.076
LVEF, %	61.8 ± 9.7	59.5 ± 9.8	64.4 ± 9.2	0.033
Tricuspid regurgitation, %	31 (45.6%)	21 (50%)	10 (32.2%)	0.095
Diastolic dysfunction, %	29 (42.6%)	20 (47.6%)	9 (29%)	0.10

Bold values signify the comparisons with statistically significant p values.

LVEF: left ventricle ejection fraction.

with the distance walked in the 6MWT, as suggested in the linear regression model ($r = 0.24$, $p < 0.01$) (Fig. 1). Patients with high utilisation of resources presented worse HAD-A and HAD-D scores, and showed significantly higher rates of probable presence of depression (31.2% versus 16.6%; $p = 0.046$) as shown in Fig. 2.

The multivariate analysis carried out applying logistic regression in significant variables showed a worse SpO₂, worse LCADL score, higher levels of peripheral blood leukocytes and high levels of fibrinogen, with an AUC of

0.829 (Table 4) to be independent determinants of high utilisation of healthcare resources. These determinants were found to be better than the BODE index for discriminating patients with high resource utilisation (Fig. 3).

Discussion

The results obtained in this study show that patients presenting high healthcare resource utilisation are

Table 3 Health-related quality of life (HRQoL), physical activity, and anxiety and depression scales.

	Global (n = 115)	High resource utilisation (n = 64)	Control group (n = 51)	p-Value
<i>Euro-QOL-5D</i>				
EQ-5D index	0.72 ± 0.31	0.64 ± 0.35	0.82 ± 0.22	0.002
EQ-5D VAS score	58.6 ± 20.1	56 ± 19	61.9 ± 21.2	0.12
EQ-5D VAS index	0.70 ± 0.25	0.64 ± 0.27	0.78 ± 0.20	0.002
<i>London Chest Activity Daily Living (LCADL) Scale</i>				
Self-care	10.5 ± 3.0	11.4 ± 3.4	9.3 ± 1.9	0.000
Domestic	17.8 ± 10.4	19.2 ± 10.4	16.2 ± 10.4	0.130
Physical	6.8 ± 1.8	7.3 ± 1.8	6.0 ± 1.5	0.000
Leisure	7.7 ± 4.4	7.5 ± 2.0	7.9 ± 6.3	0.623
Global	43.5 ± 13.8	47.1 ± 12.4	39 ± 14.1	0.002
<i>How much does your shortness of breath affect your life?</i>				
Much	33 (28.7%)	26 (40.6%)	7 (14.0%)	0.001
A little	67 (58.2%)	35 (52.2%)	32 (64.0%)	
Nothing	15 (13.1%)	3 (4.7%)	12 (22.0%)	
<i>Hospital Anxiety and Depression Scale (HADS)</i>				
Anxiety score	5.1 ± 4.0	5.8 ± 4	4.1 ± 4	0.021
Depression score	5.0 ± 3.9	6 ± 4.1	3.8 ± 2.6	0.003
Anxiety likely/definitively (%)	13 (11.3%)	10 (15.62%)	3 (6.25%)	0.216
Depression likely/definitively (%)	28 (24.3%)	20 (31.25%)	8 (16.6%)	0.046

EQ-5D index: European quality of life (EuroQOL) five-dimension total score.

Bold values signify the comparisons with statistically significant p values.

EQ-5D VAS score: European quality of life visual analogue scale.

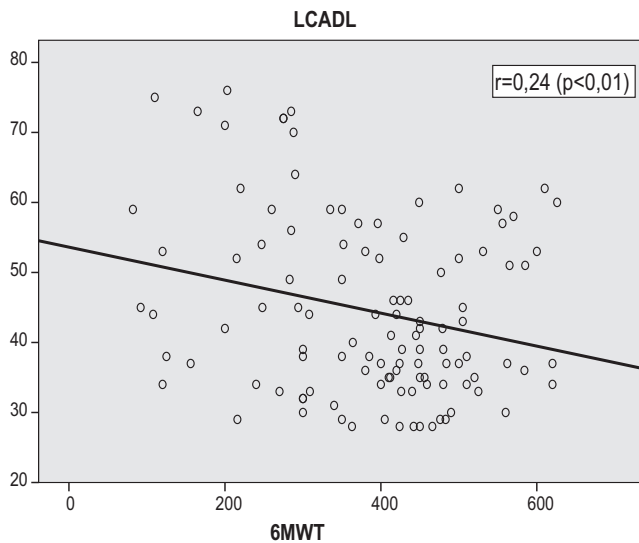


Figure 1 Relationship between the LCADL score and distance walked in the 6MWT.

characterised by lower levels of physical activity, as measured by the LCADL scale, worse SpO_2 and evidence of increased systemic inflammation, as suggested by increased fibrinogen values and greater leukocyte counts in peripheral blood. These three factors allow more effective identification of the patients with a high utilisation of healthcare resources than the BODE index, yielding an AUC of 0.829. Although validation by a prospective cohort study should be performed, the aforementioned factors may be a useful tool for identifying this subgroup of patients that place a high economic burden on healthcare systems and they can be applied at all levels of healthcare.

These results confirm the impact that reduced physical activity has on COPD patients.²¹ It is widely known that lower levels and intensity of daily physical activity can be observed in early stages of COPD.²² Limited physical activity is associated with a poorer health-related quality of life,^{23,24} a greater risk of hospital admissions,²⁵ and even higher mortality rates.^{25,26} Likewise, in our study the

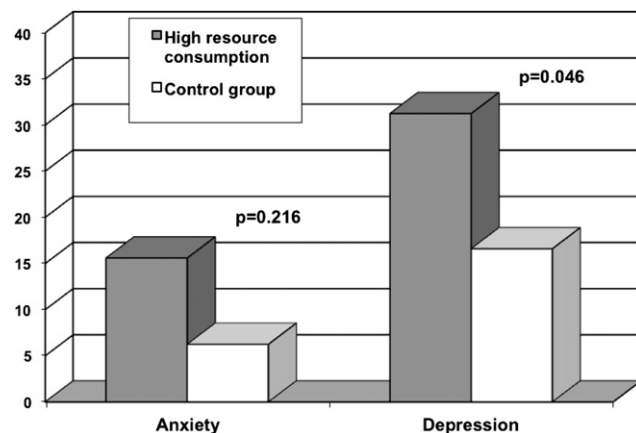


Figure 2 Anxiety and depression percentages across the study groups.

Table 4 Multivariate analysis. Determinants of high resource utilisation.

Parameter	OR	CI 95%
BODE index	0.832	0.625–1.130
SpO_2	0.869	0.760–0.994
LCADL	1.046	1.001–1.093
EQ-5D	0.548	0.060–4.999
HAD-anxiety	0.934	0.792–1.101
HAD-depression	1.111	0.926–1.334
Leukocyte count ($10^3/\mu\text{L}$)	1.295	1.032–1.625
Fibrinogen (mg/dL)	1.005	1.002–1.008
Sensitivity	67.95%	
Specificity	81.78%	
Positive predictive value (PPV)	82.73%	
Negative predictive value (NPV)	65.66%	
ROC curve, AUC	0.829	

Bold values signify the comparisons with statistically significant p values.

SpO_2 : Haemoglobin, oxygen saturation; LCADL: London Chest Activities of Daily Living Scale; EQ-5D: European quality of life (EuroQOL) five-dimension scale; HADS: Hospital Anxiety and Depression Scale; ROC curve: receiver operating characteristics curve; AUC: area under the curve.

patients with high resource utilisation also displayed significantly limited physical activity, as measured by the LCADL questionnaire and the 6MWT. In the multivariate analysis, only the LCADL score had a significant association with high use of healthcare resources. This finding may suggest that the impact of COPD on physical activity does not only become apparent in reduced physical capacity but also in other intervening psychosocial factors that increase disease impact and can be objectively measured with a standardised questionnaire.^{18,19,27} In this respect, there was a significant, albeit weak, correlation between the 6MWT and the LCADL scale. The results obtained are of great interest, since performing the 6MWT during primary care consultations is difficult to achieve. However,

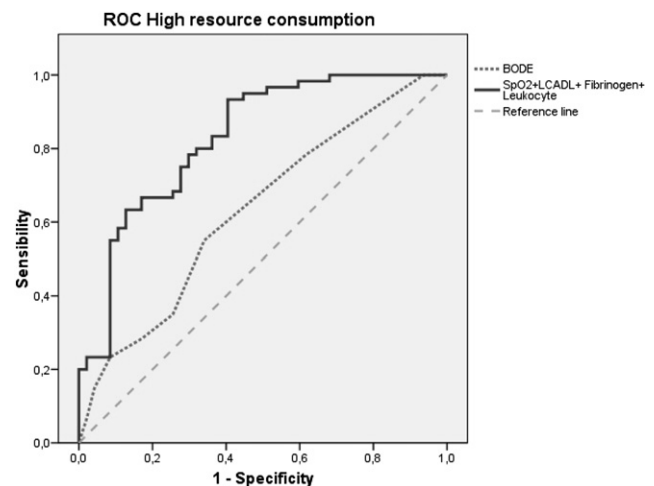


Figure 3 ROC curves for the prediction of high resource utilisation for BODE index and the proposed model (SpO_2 , leukocyte count, fibrinogen, LCADL).

administering daily physical activity questionnaires such as the LCADL, which are brief, self-completed scales, can provide useful information and are even of better predictive value in relation to the prognosis of the patient. The design of our study did not allow determination as to whether the changes in daily physical activity translate into a reduction in the utilisation of healthcare resources, although some studies have suggested this possibility. In fact, Esteban et al.²³ demonstrated that an increase in the level of physical activity is associated with a significant improvement in health status. Surprisingly, despite such evidence, there were no differences regarding the utilisation of a rehabilitation program or in pharmacological therapy across the two groups of patients.

The relationship between physical activity and quality of life was also reflected in our population. Patients with COPD and high utilisation of resources presented worse EQ-5D scores. A similar relationship was described between greater worsening of quality of life and increased utilisation of healthcare resources in a previous case-control study using the St George's Respiratory Questionnaire.¹¹ This association may be due to the relationship between acute exacerbations and hospital admissions, which is characteristic among patients with high utilisation of healthcare resources, and diminished physical activity²⁸ and quality of life.^{29,30}

There was also a clear trend towards higher levels of depression and anxiety among patients with high utilisation of resources. These patients showed a higher HAD score and a higher percentage of patients showed a probable presence depression. Depression is a common comorbidity in COPD patients,³¹ and comorbid depression has been proven to be a predictor of acute exacerbations, hospital admissions³² and greater mortality.³³ Detection of depression might enable pharmacological and psychological interventions, which might therefore potentially reduce high healthcare resource use. Previous studies highlight that adequate pulmonary rehabilitation can reduce anxiety and depression in COPD patients, and thus lead to improved quality of life and physical activity.^{34–36} Anxiety, which was more common in the study patients with high utilisation of resources, is also associated with a higher risk of acute exacerbations and hospitalisations among COPD patients.^{8,36,37} Despite the significant differences in the prevalence of anxiety and depression across the study groups in our study, these variables were not statistically significant in the multivariate analysis possibly due to an association with the LCADL score.

Interestingly, no significant differences in pharmacological treatment were observed between groups. Poor compliance with treatment might play a role in increasing healthcare resource utilisation, but the design of the study did not allow analysis of this factor.

Another observation is that the group of patients with high resource utilisation displayed greater systemic inflammation, as measured by leukocytes and fibrinogen, with respect to the control group. This result is similar to the findings of recent studies, in which exacerbations are associated with high fibrinogen levels, and are predictors of COPD-related hospital admissions and mortality.^{38–40} Moreover, the hypoxia found among high users of resources is also associated with systemic inflammation,⁴¹

as shown in our study. The relationship between greater systemic inflammation and a lower level of physical activity has been observed among other COPD patients.^{42,43} These findings suggest that patients with a high utilisation of healthcare resources and frequent acute exacerbations have reduced physical activity, elevated systemic inflammation, poor quality of life and higher rates of anxiety and depression, all of which lead to a higher risk of acute exacerbations, hence a vicious circle which is difficult to break.

It is worth pointing out that even though the sample population that underwent the echocardiographic test was small, patients with high utilisation of resources displayed a significantly reduced left ventricle ejection fraction and a clear trend towards greater pulmonary hypertension. These findings also support those of recent investigations which highlight the importance of cardiovascular comorbidity as a predictor of exacerbations,⁴⁴ and greater cardiac comorbidity in patients requiring frequent hospital care.¹¹ Further research into the relationship between cardiac comorbidity and utilisation of health resources in a larger sample of patients with COPD is required.

Given its cross-sectional nature, this study presents some limitations, which do not allow causal relations to be established. Therefore, we believe that our outcomes should be confirmed by further prospective long-term follow-up studies. Likewise, the impossibility of performing echocardiography in the whole study population poses some limitation to our conclusions on cardiac comorbidity among such patients. However, the present study has important strengths, such as its multicentre design and the inclusion of the data collected that are easy to apply in everyday clinical practice and enable its implementation in the routine management of COPD patients. The findings presented in the study suggest the presence of a group of COPD patients who place a high economic burden on health systems and present a series of distinctive characteristics. The next step would involve the study of how management of the factors linked to high utilisation of health resources can improve the prognosis of COPD patients.

Conflict of interest statement

The authors declare that they have no conflict of interest related to this manuscript.

Acknowledgement

The research group InEPOC (Innovación en EPOC) is funded by an unrestricted grant from GlaxoSmithKline.

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